

Text Normalization Pipeline for Social Media Content

A Project in Computational Linguistics

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Contents

1 Introduction:

2 Motivation:

3 Proposed Solution:

3.1 Spelling Correction:

3.2 Transliteration Variations:

3.3 Code-Switching and Code-Mixing Detection:

3.4 Abbreviations and Slang Expansion:

3.5 Emoji and Punctuation Normalization:

4 Pipeline Implementation:

5 Optional enhancements:

6 Evaluation Approach:

6.1 Dataset and Procedure:

6.2 Metrics:

7 Expected Deliverables:

8 Conclusion:

1 Introduction:

Social media and informal online platforms feature non-standard spellings, transliteration inconsistencies, abbreviations, slang, and code-switched text. Many existing NLP systems struggle to handle such noisy inputs, particularly in Indian languages. This project focuses on building a pipeline for normalizing social media text in Hindi, Tamil, and Malayalam, addressing both native-script and Roman-script inputs to improve downstream NLP tasks such as sentiment classification.

2 Motivation:

- Large volumes of user-generated text on platforms like X, Reddit, Instagram, and YouTube are written informally with non-standard or inconsistent language.
- Language models fail to handle these inputs properly due to spelling errors, transliteration variations, code-switching, and slang.
- By normalizing these inputs, downstream NLP systems receive cleaner text and show improved performance and reliability.

3 Proposed Solution:

The normalization pipeline consists of several key modules:

3.1 Spelling Correction:

- Use edit distance to detect and correct misspellings.
- Maintain a predefined dictionary of valid words.
- Incorporate phonetic matching for Romanized text.
- Perform frequency-based corrections using bigram/trigram analysis.

3.2 Transliteration Variations:

- Users often write Indian languages in Roman script (e.g., “namaste”, “vanakkam”).
- Utilize AI4Bharat or Indic NLP Library APIs to convert Roman script to Hindi (Devanagari), Tamil, or Malayalam.
- Handle multiple valid transliterations with language model scoring or context-based disambiguation.
- Maintain a fallback mapping of common transliteration variants.

3.3 Code-Switching and Code-Mixing Detection:

- Identify when users switch between English and an Indian language in the same sentence.
- Apply word-level language identification or dictionaries.
- Label code-switched segments for more targeted processing.

3.4 Abbreviations and Slang Expansion:

- Build a dictionary of social media abbreviations/slang in both Roman and native scripts.
- Use regex-based detection and replacements for common chat-style words.
- Optionally leverage context-aware models for ambiguous expansions.

3.5 Emoji and Punctuation Normalization:

- Convert emojis to standardized tokens (e.g., the smiley emoticon get converted to '`<happy_face>`').
- Replace repeated punctuation (e.g., “!!!!”) with single symbols or an emotion marker.
- Normalize elongated words (e.g., “soooo” → “so”).

4 Pipeline Implementation:

A general sequence for the normalization pipeline:

1. **Spelling Correction**
2. **Transliteration**
3. **Code-Switch Detection**
4. **Slang & Abbreviation Expansion**
5. **Emoji & Punctuation Normalization**

5 Optional enhancements:

- Train a small mBERT model on synthetic noisy-to-clean text pairs.
- Integrate with existing libraries (Indic NLP, AI4Bharat) for transliteration & language detection.

6 Evaluation Approach:

We propose measuring the efficacy of normalization by its impact on a downstream sentiment classification task.

6.1 Dataset and Procedure:

- Collect 100–300 manually annotated social media posts in Hindi, Tamil, and Malayalam.
- Feed both raw and normalized text into an existing sentiment classifier (e.g., TextBlob, IndicBERT).

6.2 Metrics:

- **Accuracy Improvement:** Compare classifier accuracy before and after normalization using a gold-labeled set.
- **Confidence Shift:** Evaluate changes in confidence scores.
- **Error Reduction Rate:** Examine how many misclassifications the normalization fixes.

7 Expected Deliverables:

- Annotated corpus of raw vs. normalized social media text.
- Source code for the normalization pipeline.
- Lexicons/dictionaries of common variations and standard forms.
- Evaluation report (quantitative + qualitative) on normalization’s impact.
- Analysis of challenges and insights for each language.

8 Conclusion:

By addressing spelling, transliteration, code-switching, slang, and emoji/punctuation usage, this project provides a robust normalization pipeline for Hindi, Tamil, and Malayalam social media text. Empirical evaluation through sentiment classification demonstrates how such normalization can substantially enhance the reliability and accuracy of downstream NLP tasks.